

AL AMRU BIL MARUF

Software Engineer

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Dhaka, Bangladesh mortytheplatypus.github.io

SKILLS

Languages: Java, JavaScript, Python, C/C++, Bash
Frameworks: Spring Boot, React, Next.js, Express.js, Flask
Databases: MySQL, PostgreSQL, MongoDB, Cassandra
Infrastructure: Nginx, Docker, Kafka, Redis, CDN, RTMP, HLS, Coroot
Tools/Protocols: Git, Liquibase, Locust, WebSocket, JWT

EXPERIENCE

June 2024
onwards

Synesis IT PLC

Software Engineer

Core contributor to Convay, a high-scale video conferencing platform.

- **Infrastructure Optimization:** Contributed to an HLS edge-caching strategy that resolved critical bottlenecks where 32-core CPUs were maxing out at 1Gbps; scaled delivery to 10k+ concurrent viewers with near-zero increase in origin load.
- **Core Features:** Developed JWT-based authentication and authorization, RBAC, meeting scheduling logic, host-group messaging for private moderator communication, and screen-share video overlay functionality allowing presenters' camera feeds to remain visible during active screen sharing.
- **Streaming Infrastructure:** Secured HLS playlist access through token-authenticated edge requests and integrated authenticated streaming workflows with Convay Big Meeting. Built RTMP proxy layer with automatic origin failover for resilient live stream ingestion, and implemented concurrent recording support on streaming origins without interrupting active broadcasts.
- **Load Testing:** Designed and developed Locust-based simulations of real browser HLS behaviour, including manifest polling and segment downloads, successfully validating streaming performance for 10k+ concurrent viewers.
- **Observability & Deployment:** Utilized monitoring tools like Coroot to monitor system health and diagnose production performance regressions. Also managed deployment and maintenance of services in development environments using Docker, systemd services, and Nginx reverse proxies.

Tech Stack: Spring Boot, React, Kafka, Docker, Nginx, Cassandra, MySQL, Liquibase, Locust, Coroot, RTMP

December 2023 –
May 2024

Sentien IO

Frontend Engineer

Led frontend development and UI/UX design for this AI-driven interview preparation platform.

- **Product & UX Collaboration:** Worked closely with leadership on product flows, UX decisions, and interface design across core platform experiences.
- **Core User Journey:** Developed multi-step onboarding and interview preparation flows covering interviewer selection, resume uploads, and targeted role tracking.
- **Dashboards & Analytics:** Built interactive user portals for saved job management, interview history tracking, and visualization of post-interview feedback through dynamic charts and gamified metrics.
- **Monitoring:** Integrated monitoring tools like Sentry for frontend error tracking and production issue monitoring.

Tech Stack: React, Next.js, TypeScript, Tailwind CSS, Sentry

May 2023 –
June 2023

MYSOFT LIMITED

Health Analytics Intern

Built a real-time health analytics dashboard to visualize hospital data trends.

- Aggregated and explored multi-dimensional patient records to surface regional and seasonal disease patterns by demographics (age, gender, location).
- Delivered a Flask and Angular full-stack app with PostgreSQL-backed reporting so administrators could monitor workload and outbreak signals from a single interface.

Tech Stack: Flask, Angular, PostgreSQL

PROJECTS

PostgreSQL
Spring Boot

JWT Authenticator

GitHub

Reusable Spring Boot + PostgreSQL auth starter: registration/login, JWT access sessions, and patterns intended for drop-in use in new services (not a demo-only scaffold).

Next.js Express.js	Find My Apartment Built an apartment-search platform with advanced filtering, personalized recommendation logic, and integrated map views for enhanced user experience.	Backend Frontend
Flask React	Pasteruf Engineered a dual-purpose platform for persistent text storage and URL shortening. Features automatic redirection logic and a streamlined interface for sharing content across platforms.	GitHub
PL/SQL React Express.js	Fantasy Premier League Fantasy-football style league app: squads, transfers, and competition flows with an Oracle/PL/SQL data layer and a React + Express.js client/API.	GitHub

EDUCATION

2019 — 2024	Bangladesh University of Engineering and Technology CGPA: 3.48/4.00 Thesis: Bangla Document Digitization using CNN	BSc in CSE
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OTHERS

- Fluent in Bengali and English
- Senior member and mentor, Dinajpur Math Club: workshops, contests, and peer coaching for secondary students